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# CSA1306

## CO1 - Assignment 1

TASK - 1 : Mathematical formulation of formal languages and Automata

Alphabet → Finite non-empty set of symbols used to construct strings. Ex : $\Sigma = \{0, 1\}$

Strings → Finite sequence of symbols taken from alphabet.

Formal Language → Collection of strings formed using a particular alphabet.

| Finite string | Empty string | Language |
|---|---|---|
| Sequence containing finite no of symbols | String containing no symbols | Set of strings over an alphabet |

Finite Automaton :-

represented using 5-tuple format

$M = (Q, \Sigma, \delta, q_0, F)$

Where,

$Q \longrightarrow$ Finite set of states

$\Sigma \longrightarrow$ Input alphabet

$\delta \longrightarrow$ Transition function

$q_0 \longrightarrow$ Initial state

$F \longrightarrow$ Set of final states

## Sample Binary strings & language membership

Let $L = \{ w \in \{0,1\}^* \mid \text{first symbol} = \text{last symbol} \}$

0 $\longrightarrow$ ✓ Accepted

1 $\longrightarrow$ ✓ Accepted

010 $\longrightarrow$ ✓ Accepted

110 $\longrightarrow$ X Rejected

1001 $\longrightarrow$ ✓ Accepted

## TASK - 2 : Mathematical design & construction of DFA

$M = (Q, \Sigma, \delta, q_0, F)$

where, $Q = \{ q_s, q_{00}, q_{01}, q_{11}, q_{10} \}$

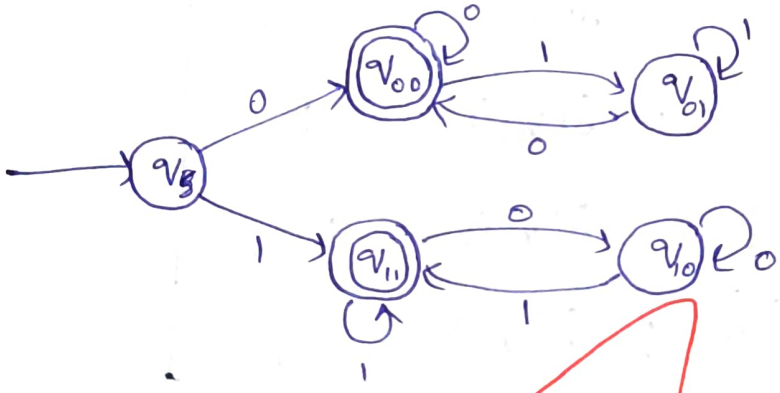
$\Sigma = \{0,1\}$

$$\delta = \begin{bmatrix} \delta(q_s,0) = q_{00} \; ; \; \delta(q_s,1) = q_{11} \\ \delta(q_{00},0) = q_{00} \; ; \; \delta(q_{00},1) = q_{01} \\ \delta(q_{01},0) = q_{00} \; ; \; \delta(q_{01},1) = q_{01} \\ \delta(q_{11},0) = q_{10} \; ; \; \delta(q_{11},1) = q_{11} \\ \delta(q_{10},0) = q_{10} \; ; \; \delta(q_{10},1) = q_{11} \end{bmatrix}$$

Transition table :-

| $Q/\Sigma$ | 0 | 1 |
|---|---|---|
| $\rightarrow q_S$ | $q_{00}$ | $q_{11}$ |
| $q_{01}$ | $q_{00}$ | $q_{01}$ |
| $q_{10}$ | $q_{10}$ | $q_{11}$ |
| $*q_{00}$ | $q_{00}$ | $q_{01}$ |
| $*q_{11}$ | $q_{10}$ | $q_{11}$ |

Transition diagram :-




Tracing of i/p strings :-

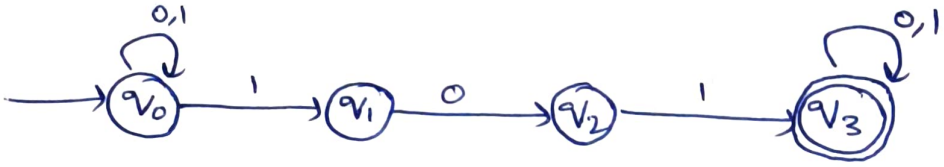
(i) 010 : $q_S \xrightarrow{0} q_{00} \xrightarrow{1} q_{01} \xrightarrow{0} q_{00}$ ✓ Accepted

(ii) 011 : $q_S \xrightarrow{0} q_{00} \xrightarrow{1} q_{01} \xrightarrow{1} q_{01}$ ✗ Rejected

(iii) 101 : $q_S \xrightarrow{1} q_{11} \xrightarrow{0} q_{10} \xrightarrow{1} q_{11}$ ✓ Accepted

(iv) 100 : $q_S \xrightarrow{1} q_{11} \xrightarrow{0} q_{10} \xrightarrow{0} q_{10}$ ✗ Rejected

TASK - 3 : Non - deterministic finite automata for pattern recognition




$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$$\delta = \begin{bmatrix} \delta(q_0,0) = \{q_0\} ; & \delta(q_0,1) = \{q_0, q_1\} \\ \delta(q_1,0) = \{q_2\} ; & \delta(q_1,1) = \phi \\ \delta(q_2,0) = \phi ; & \delta(q_2,1) = \{q_3\} \\ \delta(q_3,0) = \{q_3\} ; & \delta(q_3,1) = \{q_3\} \end{bmatrix}$$

$q_0 = \{q_0\}$

$F = \{q_3\}$

Transition table :-

| $Q \backslash \Sigma$ | 0 | 1 |
|---|---|---|
| $\rightarrow q_0$ | $\{q_0\}$ | $\{q_0, q_1\}$ |
| $q_1$ | $\{q_2\}$ | $\phi$ |
| $q_2$ | $\phi$ | $\{q_3\}$ |
| $* q_3$ | $\{q_3\}$ | $\{q_3\}$ |

Sample input :-

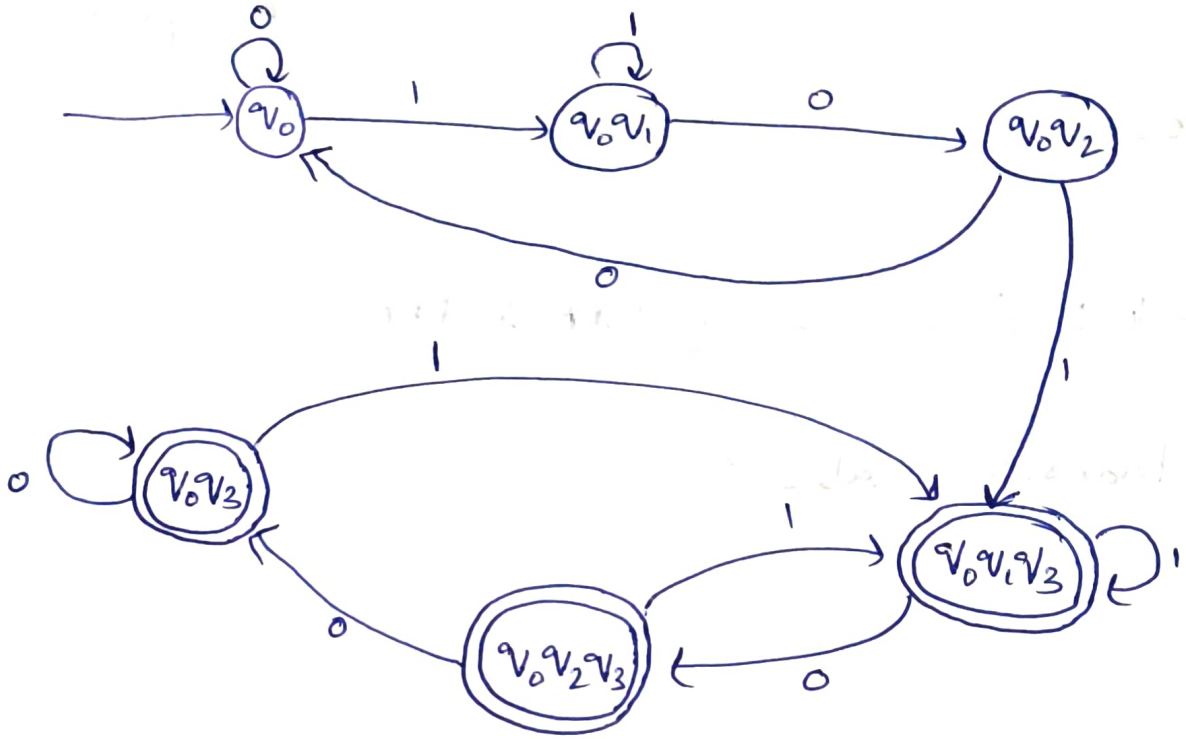
(i) 101 : $q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_3$ ✓ Accepted

(ii) 1000 : $q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0$ X Rejected

TASK - 4 : Equivalence of NFA & DFA

DFA transition table :-

| Q \ Σ | 0 | 1 |
|---|---|---|
| $\rightarrow q_0$ | $q_0$ | $[q_0 q_1]$ |
| $\rightarrow [q_0 q_1]$ | $[q_0 q_2]$ | $[q_0 q_1]$ |
| $\rightarrow [q_0 q_2]$ | $q_0$ | $[q_0 q_1 q_3]$ |
| $\overset{*}{\rightarrow} [q_0 q_1 q_3]$ | $[q_0 q_2 q_3]$ | $[q_0 q_1 q_3]$ |
| $\overset{*}{\rightarrow} [q_0 q_2 q_3]$ | $[q_0 q_3]$ | $[q_0 q_1 q_3]$ |
| $\overset{*}{\rightarrow} [q_0 q_3]$ | $[q_0 q_3]$ | $[q_0 q_1 q_3]$ |




Sample input :-

(i) 101

NFA processing : $q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_3$ ✓ Accepted

DFA processing : $q_0 \xrightarrow{1} q_0q_1 \xrightarrow{0} q_0q_2 \xrightarrow{1} q_0q_1q_3$ ✓ Accepted

(ii) 1000

NFA processing : $q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0$ ✗ Rejected

DFA processing : $q_0 \xrightarrow{1} q_0q_1 \xrightarrow{0} q_0q_2 \xrightarrow{0} q_0 \xrightarrow{0} q_0$ ✗ Rejected

# TASK-5 : ε-NFA Construction, ε-closure & Equivalence analysis

$M = (Q, \Sigma, \delta, q_0, F)$

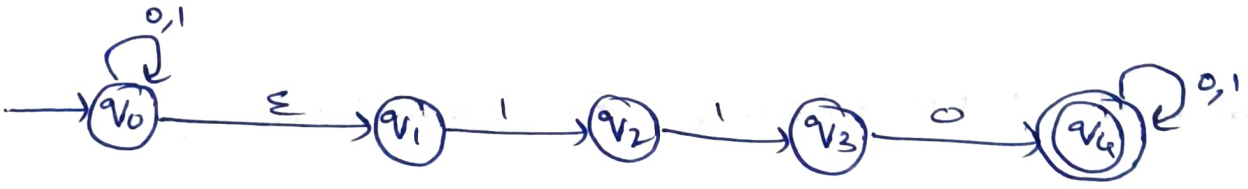
where, $Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

$q_0 = \{q_0\}$

$F = \{q_4\}$

| State | 0 | 1 | ε |
|---|---|---|---|
| → $q_0$ | $q_0$ | $q_0$ | $q_1$ |
| $q_1$ | $\phi$ | $q_2$ | $\phi$ |
| $q_2$ | $\phi$ | $q_3$ | $\phi$ |
| $q_3$ | $q_4$ | $\phi$ | $\phi$ |
| * $q_4$ | $q_4$ | $q_4$ | $\phi$ |




E - closure :-

$E(q_0) \longrightarrow q_0, q_1$

$E(q_1) \longrightarrow q_1$

$E(q_2) \longrightarrow q_2$

$E(q_3) \longrightarrow q_3$

$E(q_4) \longrightarrow q_4$

$\delta'(q_0, 0) = E(\delta(q_0, 0) \cup \delta(q_1, 0)) = \{q_0, q_1\}$

$\delta'(q_0, 1) = E(\delta(q_0, 1) \cup \delta(q_1, 1)) = \{q_0, q_1, q_2\}$

$\delta'(q_1, 0) = E(\phi) = \phi$

$\delta'(q_1, 1) = E(q_2) = \{q_2\}$

$\delta'(q_2, 0) = E(\phi) = \phi$

$\delta'(q_2, 1) = E(q_3) = \{q_3\}$

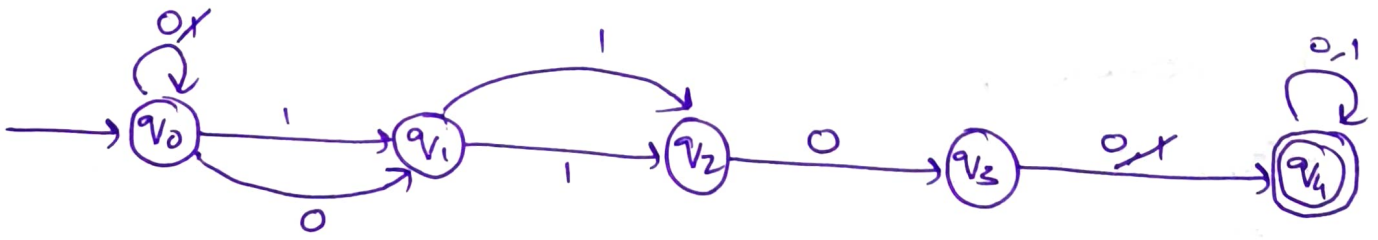
$\delta'(q_3, 0) = E(q_4) = \{q_4\}$

$\delta'(q_3, 1) = E(\phi) = \phi$

$\delta'(q_4, 0) = E(q_4) = \{q_4\}$

$\delta'(q_4, 1) = E(q_4) = \{q_4\}$

Equivalent NFA without $\varepsilon$- transitions :-



| State | 0 | 1 |
|---|---|---|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_1, q_2\}$ |
| $q_1$ | $\phi$ | $\{q_2\}$ |
| $q_2$ | $\phi$ | $\{q_3\}$ |
| $q_3$ | $\{q_4\}$ | $\phi$ |
| $*q_4$ | $\{q_4\}$ | $\{q_4\}$ |

Verification :-

110 ✓ Accepted

0110 ✓ Accepted

1010 ✗ Rejected

1111 ✗ Rejected

4/5/20